



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 9 • STANDARD 6.NOS.9

# Distance on the Coordinate Plane

Lesson 9-6 • Teacher Copy

**Name:** \_\_\_\_\_ **Date:** \_\_\_\_\_ **Class:** \_\_\_\_\_

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## TODAY'S OBJECTIVES

**Content Objective:** I can find the distance between two points on the coordinate plane using absolute value.

**Language Objective:** I can explain my work using the words distance, absolute value, horizontal distance, and vertical distance.

See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



## WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Distance

Absolute value

Horizontal distance

Vertical distance

Coordinate plane

Integer



Tap any word to see what it means and a picture.

1

How far apart two points are on the coordinate plane is the

.

2

The distance of a number from zero, used to find distance between points, is its

.

3

The distance between two points along the x-direction is the

.

4

The distance between two points along the y-direction is the

.

5

The grid made by the x-axis and y-axis is the

.

6

A whole number or its opposite is an  .

## Reflect — Exit Ticket

What is the distance between the points (3, -2) and (3, 5)?

- A. 3 units
- B. 5 units
- C. 7 units
- D. 9 units

Your answer:

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## Answer Key & Teacher Guide

1. **Notes 1:** Distance
2. **Notes 2:** Absolute value
3. **Notes 3:** Horizontal distance
4. **Notes 4:** Vertical distance
5. **Notes 5:** Coordinate plane
6. **Notes 6:** Integer
7. **Watch for this mistake:** *A common mistake in Distance on the Coordinate Plane is skipping the key idea: "If points are on the same side of zero, subtract; if they are on opposite sides of zero, add the absolute values." — always check your work against this rule before you submit.*
8. **Try It 1:** B. 7 blocks — *Same  $y = 5$ , so the segment is horizontal. The points are on opposite sides of zero, so add the absolute values:  $|-4| + |3| = 4 + 3 = 7$  blocks.*
9. **Try It 2:** B. 5 blocks — *Same  $x = -2$ , so the segment is vertical. The points are on opposite sides of zero:  $|1| + |-4| = 1 + 4 = 5$  blocks.*
10. **Exit Ticket:** C. 7 units — *Both points share  $x = 3$ , so this is a vertical distance.  $|5 - (-2)| = |5 + 2| = 7$  units.*